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PyPEEC: A 3D Quasi-Magnetostatic Solver using an FFT-Accelerated PEEC Method with Voxelization

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Summary

The partial element equivalent circuit method (PEEC) is a particularly interesting integral-equation method for electromagnetic problems as it can be easily combined with circuit simulations and does not require a discretization of the free space. This paper introduces PyPEEC, an open-source Python solver targeting 3D quasi-magnetostatic problems such as inductors, transformers, chokes, IPT coils, or busbars. The geometry is described with voxels and the solver uses an FFT-accelerated variant of the PEEC method. The FFT acceleration drastically reduces the computational costs and the memory footprint.

Statement of Need

Quasi-magnetostatic field simulations are widely used for the design and optimization of electrical components (e.g., power electronics, packaging, IC design). Among the existing numerical methods (e.g., FEM, FDTD, PEEC, BEM), the PEEC method features several advantages (Jithesh & Pande, 2003; A. Ruehli et al., 2017; A. E. Ruehli, 1974; Torchio, 2019):

- Only the active materials are discretized (no need to mesh the vacuum/air).
- Intuitive understanding of the equation discretization process.
- Straightforward connection of external circuit elements.

The fundamental drawback of the classical PEEC method is that the matrix describing the equation system is not sparse. This means that the computational cost and the memory requirement become problematic for large problems.

This problem can be mitigated if the geometry is represented with voxels (Torchio, Lucchini, et al., 2022; A. C. Yucel et al., 2018). The first advantage of such geometries is that the Green's functions (i.e. the inductance and potential matrices) have analytical solutions, avoiding computationally intensive numerical integrals (Hoer & Love, 1965; Piatek & Baron, 2012). The second advantage is that, due to the regular voxel discretization, many coefficients of the inductance and potential matrices are redundant. The dense matrices are block-Toeplitz Toeplitz-block matrices and feature the following key properties (Lee, 1986; Polimeridis et al., 2014):

- The block-Toeplitz Toeplitz-block matrices can be embedded into circulant tensors reducing the memory requirements from $O(n^2)$ to $O(n)$.
- The matrix-vector multiplications can be done with Fourier transforms. With an FFT algorithm, the computational complexity of matrix-vector multiplications is reduced from $O(n^2)$ to $O(n \log(n))$.

Different variants of the FFT-accelerated PEEC method have been shown to be extremely fast for a large variety of electromagnetic problems such as high-frequency transformers (Torchio, Lucchini, et al., 2022), power inductors (Marconato & Lucchini, 2021), human body field exposure (Torchio, Arduino, et al., 2022), nuclear fusion devices (Bettini et al., 2020), and PCB layouts (Romano et al., 2023).

An open-source implementation ("FFT-PEEC") of the FFT-accelerated PEEC method already exists (Torchio & Lucchini, 2020, 2021). However, this code can only handle magnetic materials for static simulations (and not in the frequency domain). Moreover, depending on the considered quasi-static problem (e.g. geometries with multiple conductors), the source code requires some adaptations. The open-source tools "VoxHenry" and "MARIE" are also used FFT-accelerated methods but are specialized for full-wave solutions without magnetic materials (Villena et al., 2015; A. Yucel, 2018). Finally, it should be noted that all the aforementioned implementations depend on proprietary programming languages and libraries.

PyPEEC, the tool introduced in this paper, addresses these needs by providing a 3D quasi-magnetostatic implementation of the FFT-accelerated PEEC method (Torchio, Lucchini, et al., 2022). PyPEEC is a fully open-source (MPL 2.0 licence and implemented in Python) and offers a general-purpose solver by allowing the description of arbitrary problems without having to modify the source code.

Capabilities and Workflow

In this paper, the release 5.5 of PyPEEC is considered. PyPEEC solves 3D magnetostatic and quasi-magnetostatic problems with voxel geometries. An arbitrary number of conductive domains, magnetic domains (ideal and/or lossy), and sources (voltage and/or current) can be used. The current density, flux density, electric potential, magnetic potential, and loss density are computed. The free space magnetic field (near-field) can also be computed on a point cloud. Additionally, the voltage, current, complex power, and impedance of the different terminals are extracted.

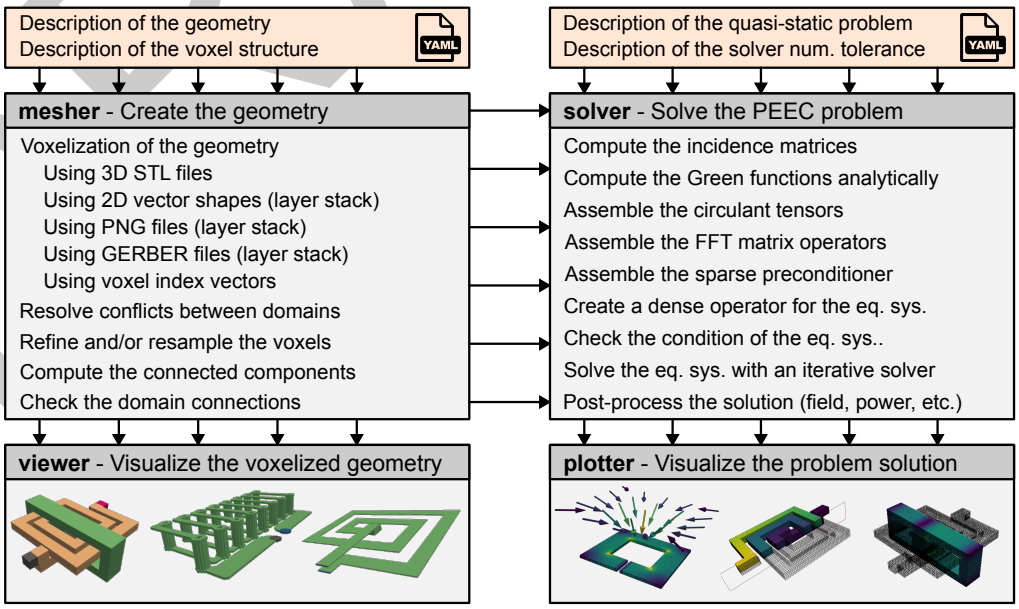

Figure 1: PyPEEC workflow consisting of the *mesher*, *solver*, *viewer*, and *plotter*

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PyPEEC is implemented in pure Python using NumPy, SciPy, Joblib, Rasterio, Shapely, Pillow, Matplotlib, and PyVista. The solver is able to leverage multi-core CPUs and GPUs. Optional HPC libraries are available for accelerating the sparse preconditioner factorization (MKL/PARDISO and PyAMG) and the FFT operations (FFTW, MKL/FFT, and CuPy/CUDA). PyPEEC can be used through an API, a command-line tool, or Jupyter notebooks. The package is available through the Python package index (PyPi) and the community-driven packaging for Conda (conda-forge). A Docker image with JupyterLab is also maintained.

Figure 1 depicts the PyPEEC workflow. First the *mesher* builds the geometry (from vector shapes, STL files, PNG files, or GERBER files), performs the voxelization, and checks the validity of the discretization. Afterward, the *solver* creates the FFT-multiplication operators, assembles the equation systems, extract sparse preconditioners, solves the problem, and post-processes the solution. The *viewer* and the *plotter* are used to visualize the results. Aternatively, ParaView can be used to analyze and visualize the solutions.

Solver Performance

To demonstrate the performance of the solver, a planar air-core spiral inductor is considered. The inductor has a footprint of 4 mm^2 and is operated in the 40.68 MHz ISM band. Figure 2 shows the ratio between the AC and DC current densities, the relative error on the extracted impedance, and the computational cost. The number of degrees of freedom represents the number of unknowns for the PEEC problem (dense equation system). The computational cost is evaluated with an AMD EPYC 7543 CPU (without GPUs).

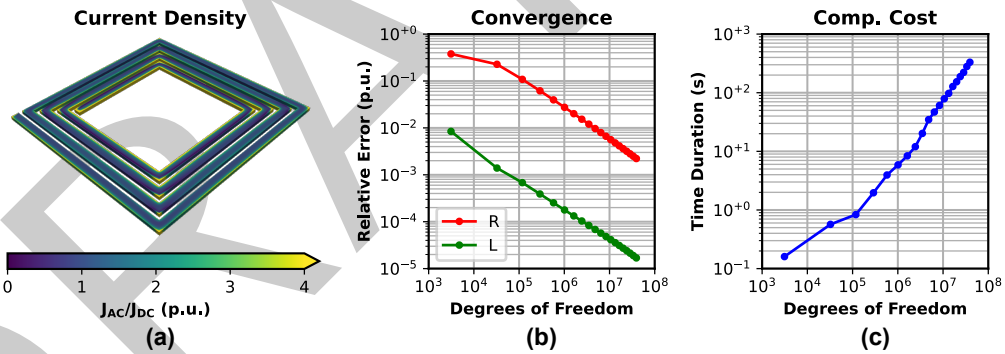

Figure 2: (a) Ratio between the AC and DC current densities. (b) Relative error on the extract equivalent resistance and the inductance. (c) Wall clock time duration for the complete workflow.

As the skin depth is smaller than the dimension of the conductor, the eddy currents are non-negligible. This implies that, with a coarse discretization, the resistance value is less accurate than the inductance value. With a tolerance of 4%, the quasi-static problem is solved in 4 seconds. The FFT acceleration allows dense PEEC problems with 10^7 degrees of freedom to be solved in 80 seconds.

The performances of PyPEEC are impacted by the following factors. For problems dominated by eddy-currents and/or containing magnetic materials, the majority of the computational effort (typically over 70%) is linked to the FFT accelerated multiplications. In such cases, the performances are directly driven by the FFT solver and benefits from the available parallel and GPU algorithms. Finally, it should be noted that PyPEEC is using a regular voxel structure to represent the geometry. This implies that large geometries with small features cannot be meshed efficiently.

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